PROJECT S.E.E.S. — SYSTEM ARCHITECTURE SPECIFICATION

ARCHITECTURE BLUEPRINT V1.0

MULTI-DOMAIN VEHICULAR ENERGY HARVESTING & AUXILIARY POWER DECOUPLING

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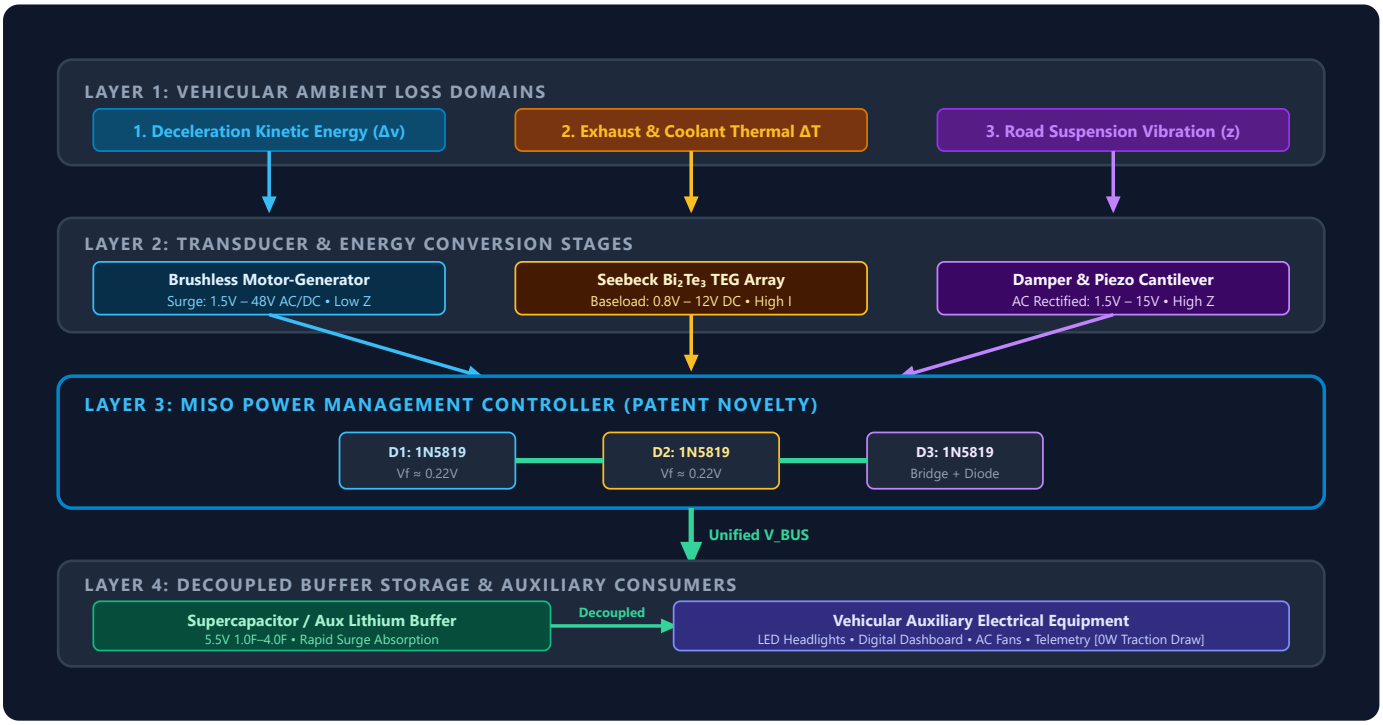
Document Control: Engineering Architecture v1.0

System Classification: Multi-Input Single-Output (MISO) Hybrid Power Bus

Application Domain: Automotive Auxiliary Load Decoupling

1. End-to-End System Energy Architecture Diagram

The S.E.E.S. architecture organizes vehicular energy recovery into four strictly decoupled physical layers, routing energy through an autonomous diode-isolated bus into a dedicated secondary buffer:



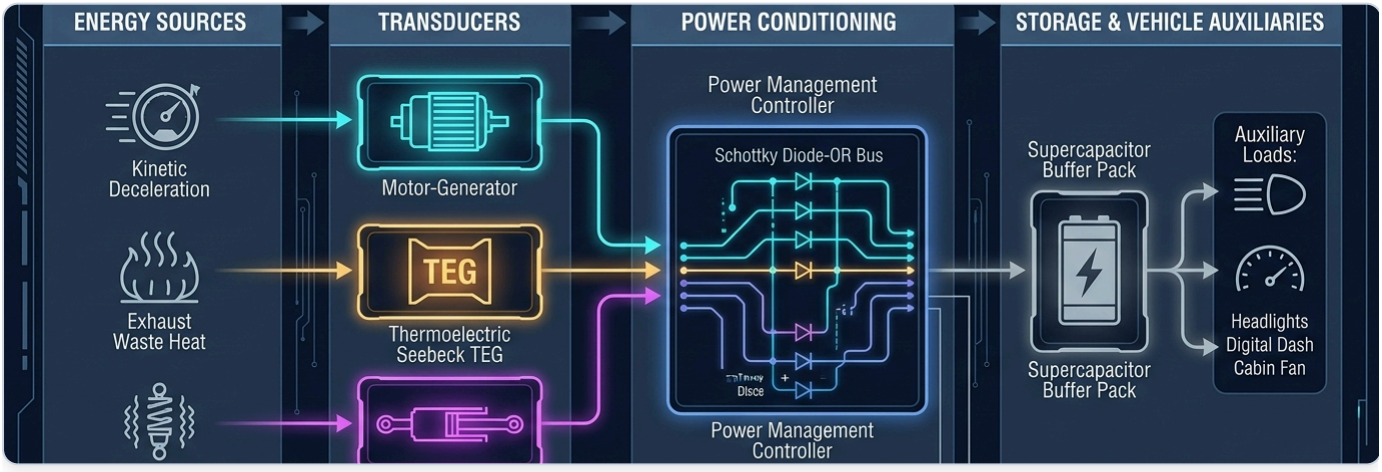


Figure 1.1: Multi-Layer Photorealistic Engineering Architecture Infographic

2. Detailed Specifications of the Four Architecture Layers

Layer 1 — Vehicular Ambient Energy Sources

Boundary: Physical Vehicle

Captures three naturally occurring ambient waste streams during operation: (1) **Kinetic Inertia** released during vehicle deceleration events, (2) **Thermal Gradient (ΔT)** between hot exhaust surfaces (200°C–400°C) and passing ambient airflow (30°C), and (3) **Mechanical Vibration** induced by road surface roughness through the vehicle unsprung mass.

Layer 2 — Energy Conversion Transducers

Domain: Mechanical-to-Electrical

Three specialized converters operate simultaneously: (1) **Permanent Magnet Motor-Generators** producing high-power charging bursts during braking, (2) **Bismuth Telluride (Bi_2Te_3) Seebeck Modules** generating continuous DC baseload current via electron diffusion, and (3) **Bimodal Dampers** pairing high-impedance piezoelectric cantilevers for micro-telemetry with electromagnetic dampers for suspension travel.

Layer 3 — MISO Power Management Controller (Patent Novelty)

Domain: Power Conditioning

The core patented innovation of S.E.E.S. is the **Multi-Input Single-Output (MISO) Diode-OR Combining Bus**:

- **Cross-Conduction Prevention:** Ultra-low forward drop Schottky diodes (1N5819, $V_f \approx 0.22\text{V}$) isolate each generator channel. When the kinetic generator spikes to 4.58V during deceleration, diode D2 reverse-biases automatically, completely protecting the 1.70V Seebeck module from over-voltage destruction.
- **Zero Digital Switching Overhead:** Eliminates power-hungry micro-controller matrix switches that typically waste 5W–15W in standby mode, allowing efficient energy accumulation even during low-speed crawling.

Layer 4 — Decoupled Storage Buffer & Auxiliary Loads

Domain: Electrical Supply

Harvested electricity charges an autonomous secondary buffer (electrochemical supercapacitor bank or auxiliary 12V LiFePO4 battery):

- **Surge Absorption & Baseload Storage:** The supercapacitor absorbs transient deceleration spikes and continuous thermoelectric baseload, maintaining stable $2.4\text{V to }12\text{V}$ bus potentials.
- **Decoupled Auxiliary Load Delivery:** Powers auxiliary consumers (LED headlights, digital instrument cluster, cooling fans, and telemetry radios) with **zero current drawn from the main vehicle traction battery**, directly extending vehicle driving range.

3. Comprehensive System Interface Matrix

Subsystem Interface	Voltage Domain	Current Characteristic	Impedance Level	Conditioning Strategy
Kinetic Channel	1.5V – 48V (AC/DC)	High Surge (100 mA – 100 A)	Low ($20\ \Omega\text{ to }0.5\ \Omega$)	Synchronous Rectification + Schottky Diode D1
Thermoelectric Channel	0.8V – 12V DC	Continuous Baseload (50 mA – 15 A)	Low-Medium ($24\ \Omega\text{ to }2\ \Omega$)	Synchronous Boost MPPT + Schottky Diode D2
Vibration Channel	1.5V – 15V AC Spikes	Micro-Current ($1\ \text{mA to }200\ \text{mA}$)	High ($450\ \Omega\text{ to }10\ \text{k}\Omega$)	Full-Wave Bridge Rectifier + Schottky Diode D3
Common Output Bus	**2.4V – 12.0V DC**	**Summed Charge Current**	**Regulated**	**Supercapacitor Storage Buffer (1.0F – 4.0F)**
Auxiliary Output	5.0V / 12.0V DC	Steady Continuous Draw	Load-Dependent	Isolated Decoupled Auxiliary Rail

4. Core Engineering Benefits of this Architecture

1. **Surge + Baseload Complementarity:** Solves the intermittency problem of regenerative braking by combining it with the continuous thermal gradient of vehicle exhaust.
2. **Electrical Immunity:** No single generator can back-feed or damage another generator under any driving condition.
3. **Autonomous Decoupling:** Auxiliary loads run completely independent of main battery state of charge, improving EV battery longevity and safety.